

Reinforcement Learning 2025/26 – LM Artificial Intelligence – University of Verona

List of papers available for presentations

Each paper can be presented by at most 2 students. The presentation lasts 40 minutes (30 mins of presentation + 10 of questions) for 2 students and 30 minutes (25 mins of presentation + 5 of questions) for a single student. Half time should be dedicated to present the paper and the other half to present the code implementing the methodology. Usually the code is available with the paper, if not please let me know and we'll find a solution.

1. **Dreamer (model-based RL, world models):** Danijar Hafner, Timothy Lillicrap, Jimmy Ba, Mohammad Norouzi. 2020. **Dream to Control: Learning Behaviors by Latent Imagination.** International Conference on Learning Representations (ICLR 2020), 1-20 (<https://arxiv.org/pdf/1912.01603>).
2. **Dreamer v3 (model-based RL, world models):** Danijar Hafner, Jurgis Pasukonis, Jimmy Ba & Timothy Lillicrap. 2025. **Mastering diverse control tasks through world models.** Nature volume 640, 647-653 (<https://www.nature.com/articles/s41586-025-08744-2>).
3. **Dreamer v4 (model-based RL, world models):** Danijar Hafner, Wilson Yan, Timothy Lillicrap. 2025. **Training Agents Inside of Scalable World Models.** Arxiv 2509.24527, 1-32 (<https://arxiv.org/pdf/2509.24527>).
4. **CQL (offline RL):** Aviral Kumar, Aurick Zhou, George Tucker, Sergey Levine. 2020. **Conservative Q-Learning for Offline Reinforcement Learning.** 34th Conference on Neural Information Processing Systems (NeurIPS 2020), 100, 1179-1191 (<https://sites.google.com/view/cql-offline-rl>).
5. **Cal-QL (offline-2-online RL):** Mitsuhiko Nakamoto, Yuexiang Zhai, Anikait Singh, Max Sobol Mark, Yi Ma, Chelsea Finn, Aviral Kumar, Sergey Levine. 2023. **Cal-QL: calibrated offline RL pre-training for efficient online fine-tuning.** 37th Conference on Neural Information Processing Systems (NeurIPS 2023), 2719, 62244 - 62269 (https://proceedings.neurips.cc/paper_files/paper/2020/file/f7efa4f864ae9b88d43527f4b14f750f-Paper.pdf).
6. **IQL (offline RL):** Ilya Kostrikov, Ashvin Nair, Sergey Levine. 2022. **Offline Reinforcement Learning with Implicit Q-Learning.** International Conference on Learning Representations (ICLR 2022) (<https://arxiv.org/pdf/2110.06169>).
7. **TD3+BC (offline RL):** Scott Fujimoto, Shixiang (Shane) Gu. 2020. **A Minimalist Approach to Offline Reinforcement Learning.** 35th Conference on Neural Information Processing Systems (NeurIPS 2021), 1540, 20132 - 20145 (https://proceedings.neurips.cc/paper_files/paper/2021/file/a8166da05c5a094f7dc03724b41886e5-Paper.pdf).
8. **(Offline-2-online RL):** Seunghyun Lee, Younggyo Seo, Kimin Lee, Pieter Abbeel, Jinwoo Shin, 2022. **Offline-to-Online Reinforcement Learning via Balanced Replay and Pessimistic Q-Ensemble.** 5th Conference on Robot Learning (CoRL 2021), 164:1702-1712 (<https://arxiv.org/pdf/2107.00591>).
9. **DT (Decision Transformers):** Lili Chen, Kevin Lu, Aravind Rajeswaran, Kimin Lee, Aditya Grover, Michael Laskin, Pieter Abbeel, Aravind Srinivas, Igor Mordatch, 2021. **Decision Transformer: Reinforcement Learning via Sequence Modeling.** Proceedings of

the 35th International Conference on Neural Information Processing Systems (NeurIPS 2021), 1156, 15084-15097(<https://sites.google.com/berkeley.edu/decision-transformer>).

10. **QT (offline RL + transformers):** Shengchao Hu, Ziqing Fan, Chaoqin Huang, Li Shen, Ya Zhang, Yanfeng Wang, Dacheng Tao, 2024. **Q-value Regularized Transformer for Offline Reinforcement Learning**. Proceedings of the 41st International Conference on Machine Learning (ICML2024), 772, 19165 - 19181 (<https://arxiv.org/pdf/2405.17098>).
11. **ODT (RL + transformers):** Qinqing Zheng, Amy Zhang, Aditya Grover, 2022. Online Decision Transformer. Proceedings of the 39 th International Conference on Machine Learning (ICML 2022), 162:27042-27059 (<https://proceedings.mlr.press/v162/zheng22c.html>).
12. **TD3+ODT (offline RL + transformers):** Kai Yan, Alexander G. Schwing, Yu-Xiong Wang, 2024. **Reinforcement Learning Gradients as Vitamin for Online Finetuning Decision Transformers**. 38th Conference on Neural Information Processing Systems (NeurIPS 2024), 1219, 38590 - 38628 (<https://arxiv.org/pdf/2410.24108>).
13. **TDQC (Calibrated VLA):** Shelly Francis-Meretzki, Mirco Mutti, Yaniv Romano, Aviv Tamar, 2020. **Temporal Difference Calibration in Sequential Tasks: Application to Vision-Language-Action Models**. Arxiv, 2604.20472 (<https://shellytechnion.github.io/TDQC.github.io/>).
14. **Moor (model-based offline RL):** Su, X., Li, P. & Chen, S. (2025) **Moor: Model-based offline policy optimization with a risk dynamics model**. Complex Intell. Syst. 11, 20 (2025). (<https://doi.org/10.1007/s40747-024-01621-x>)
15. **COMBO (model-based offline RL):** T. Yu, A. Kumar, R. Rafailov, A. Rajeswaran, S. Levine, C. Finn. (2021) COMBO: Conservative Offline Model-Based Policy Optimization. Advances in Neural Information Processing Systems 34 (NeurIPS 2021). (https://proceedings.neurips.cc/paper_files/paper/2021/file/f29a179746902e331572c483c45e5086-Paper.pdf)
16. **MACURA (model-based online RL):** B. Frauenknecht, A. Eisele, D. Subhasish, F. Solowjow, and S. Trimpe. 2024. **Trust the model where it trusts itself: model-based actor-critic with uncertainty-aware rollout adaption**. In Proceedings of the 41st International Conference on Machine Learning (ICML'24), Vol. 235. JMLR.org, Article 558, 13973–14005. (<https://raw.githubusercontent.com/mlresearch/v235/main/assets/frauenknecht24a/frauenknecht24a.pdf>)